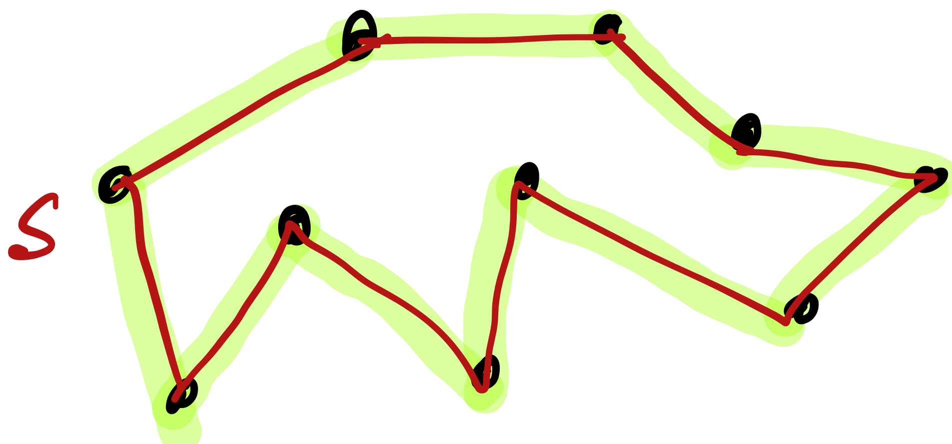
Traveling Salesman Problem (TSP)

Coping with interactability

Setup

Given a set of locations + distances betⁿ each pair

Q. What is the shortest route that visits each location exactly once & returns to the starting location.



Little History

- First formulated by William Hamilton (a.k.a — Hamiltonian cycle)

- Richard Karp showed in 1972 Hamiltonian cycle is NP-Complete

⇓

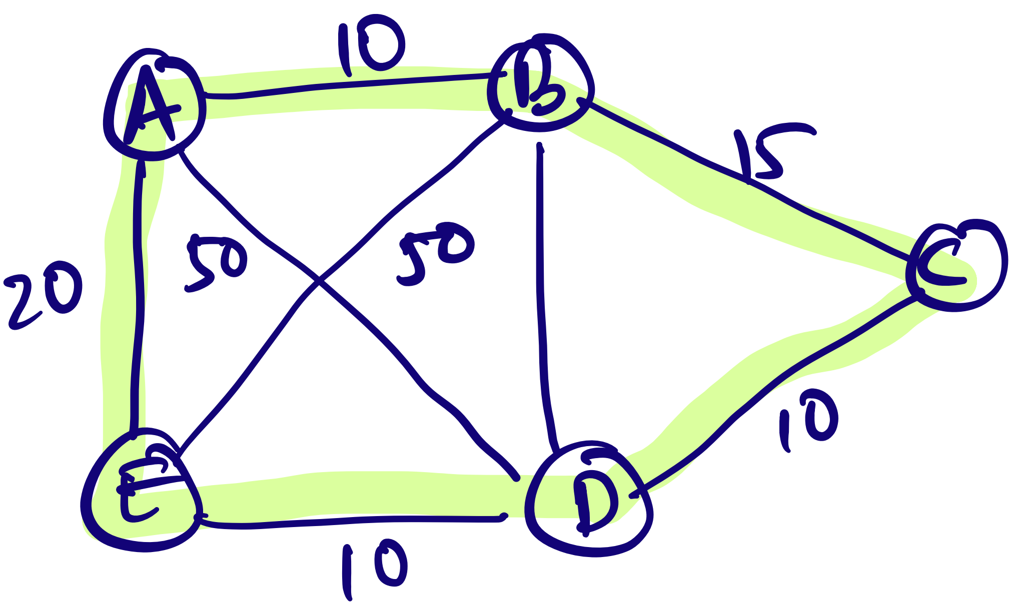
NP-Completeness for TSP

Graph problem

$G = (V, E)$ — Undirected weighted graph.

Locations are vertices;
Connections are edges.

Goal:- Minimize the sum of edge weights while visiting all nodes exactly once.



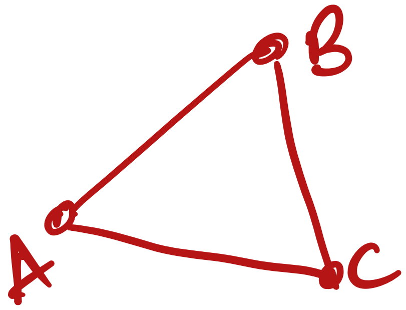
Two variants

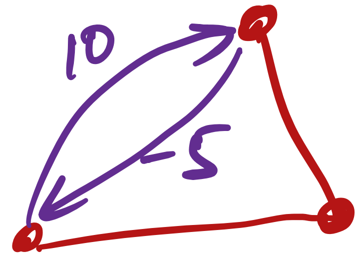
Symmetric

$|AB| = |BA|$.

Asymmetric

Directed graph.
(Much Harder prob.).





Exact Algorithms

Try all choices
(Brute-force Search)

$O(n!)$

Held-Karp DP Based
Algorithm

- $O(n^2 2^n)$

Current best known (with quantum speedup)
Exact
algo.

- $O(1.728^n)$

(SODA 2019)

Graph defn: of TSP

Tour of a graph G is a path (or, cycle) which visits all vertices & returns to its starting vertex.
a.k.a. Hamiltonian cycle)

Say, edge weights $d(\cdot)$

Sequence of vertices

$$\pi = \langle v_1, v_2, \ldots, v_n \rangle$$

Cost $$d(\pi) = \sum_{i=1}^{|\pi|-1} d(v_i, v_{i+1})$$

In-approximability

Can't be approximated in poly. time by any constant $c > 1$, unless $P = NP$.

Proof sketch

Suppose $\exists$ a c-factor apx. A for TSP.

- It means given $G = (V, E)$, A outputs a hamiltonian cycle of total wt.

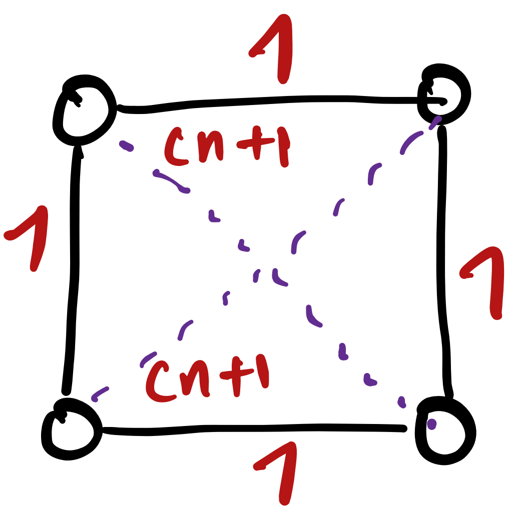
at most $c \times OPT(G)$.

- Suppose we are given an instance of the Hamiltonian cycle $G=(V,E)$
- Construct a graph $G'=(V',E')$

Complete graph on V

Wt. assignment

We assign wts. to edges



$$w(e) = \begin{cases} 1 & \text{if } e \in E. \\ cn+1 & \text{if } e \notin E. \end{cases}$$

Let's run A on G'

Since A is a c-apx.

$$Opt(G') \leq A(G') \leq c \times Opt(G')$$

Consider G:

Either G has a Hamiltonian cycle.

or it doesn't.

1. If it has a Ham. cycle.

then G' has a TSP Sol^n. of wt. n.

Since we will use only $w(e) = 1$

Hence OPT is n.

if A returns a cycle of wt. $(\leq)$ cn.

then there must be a Hamiltonian cycle.

2. If G doesn't have a Ham. cycle

then A must have used one of the edges $e \in E' \setminus E$.

and $w(e) = cn+1$.

therefore
$A(G')$ would output a Ham. cycle with total value at least $cn+1$.

greater than c-apr.

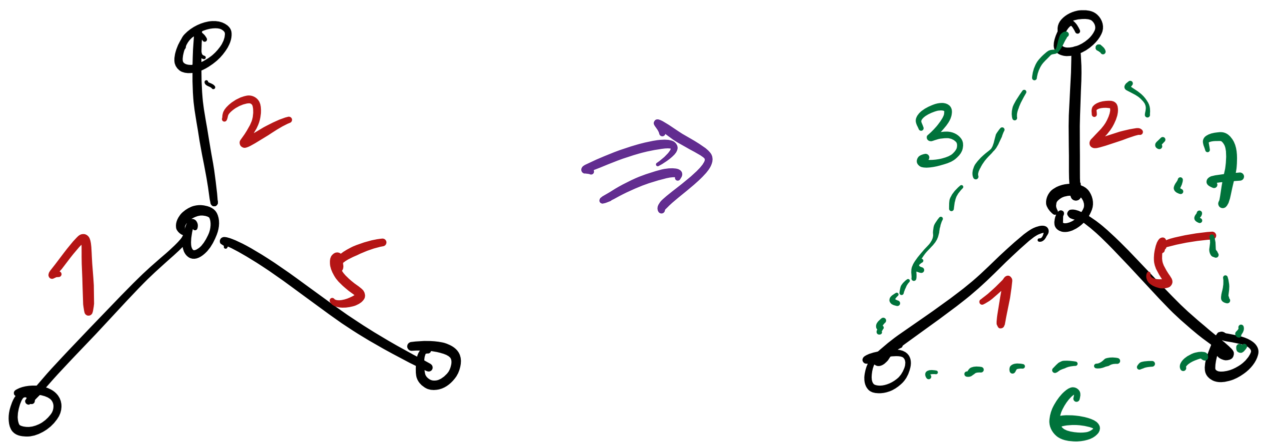
Hence to check if G has a hamiltonian cycle

Poly-time procss.

We need to check if $A(G')$ returns desired value as o/p. (Ham. cycle decision is NP-hard).

TSP on Metric Completion of G

Def^n:- Metric Completion of $G = (V,E)$ with non-negative edge wts. $d(\cdot)$ is the complete graph with edge wts. $d^*(\cdot)$



Metric Space

Tuple (V, d)

V = set of pts.

$d(\cdot)$ = dist. fn.

Satisfying

1. Non-negativity

$\forall\, v, w \in V, \quad d(v, w) \geq 0.$

2. Symmetry —

$\forall\, v, w \in V, \quad d(v, w) = d(w, v)$

3. Triangle ineq.

$\forall\ v, w, z \in V$

$d(v,w) \leq d(v,z) + d(z,w)$

Popular metrics —

Euclidean.

Doubling

planar

shortest path. etc.

Apx. of TSP via MST

Since the "tour" spans the graph optimal value of a tour is at least the value of MST

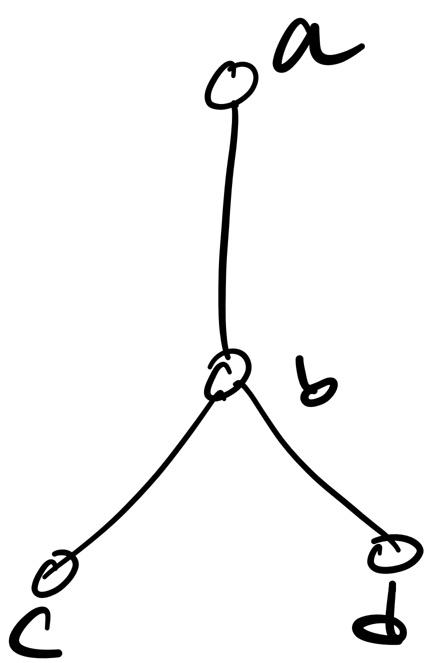
$$OPT(TSP) \geq OPT(MST)$$

Goal: Get a tour $d^*(\pi)$ s.t.

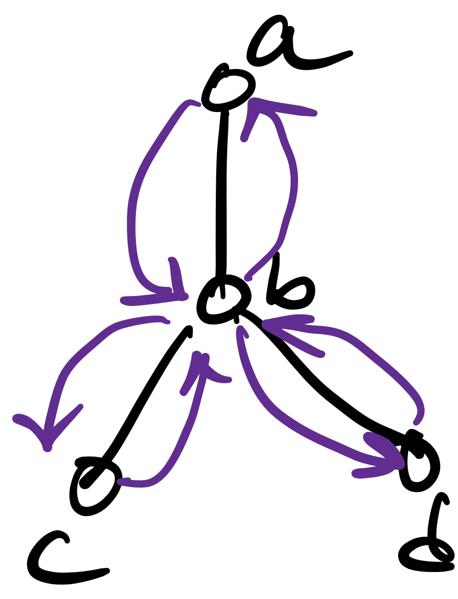
$$d^*(\pi) \leq 2 \times OPT(MST).$$

Let T be a MST of G.

- For the moment, if we ignore the Constraint of visiting each node exactly once.



Idea - In-Order traversal.



Issue :- lots of vertices repeat.

How to fix?

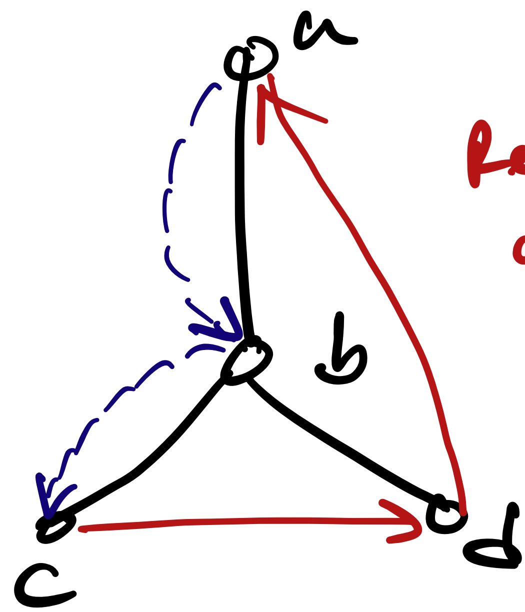
Short cutting

- Construct a tour that does not revisit the vertices

When IN-ORDER traversal repeats a vertex —
Directly travel to the next unvisited node.

(Can we do that?
Yes, G is a complete graph).

Ex.



Red edges are short-cutting.

Cost :- $d^*(\cdot)$ is metric, due to triangle ineq. — cost of taking this edge is at least as short as the original IN-ORDER path.

Hence, we get a tour π that visits each node exactly once.

Cost is at most the cost of the IN-ORDER traversal.

(Bounded above by $2 \times OPT(MST)$)

$$d^*(\pi) \leq 2 \times OPT(MST) \leq 2\, OPT(TSP)$$

This gives 2-apx.

Q. Can we do better?

- Yes.

1.5 - apx

for any metric.

(christofides alg.)

Now, we can beat slightly

$1.5 - 10^{-29}$.

(100 pages breakthrough)

Euclidean metric

PTAS

(Gödel prize winning work – Arora, Mitchell)

For general metric PTAS not possible

Care more? Stay tuned with CS 602!
(Advanced Algo.)